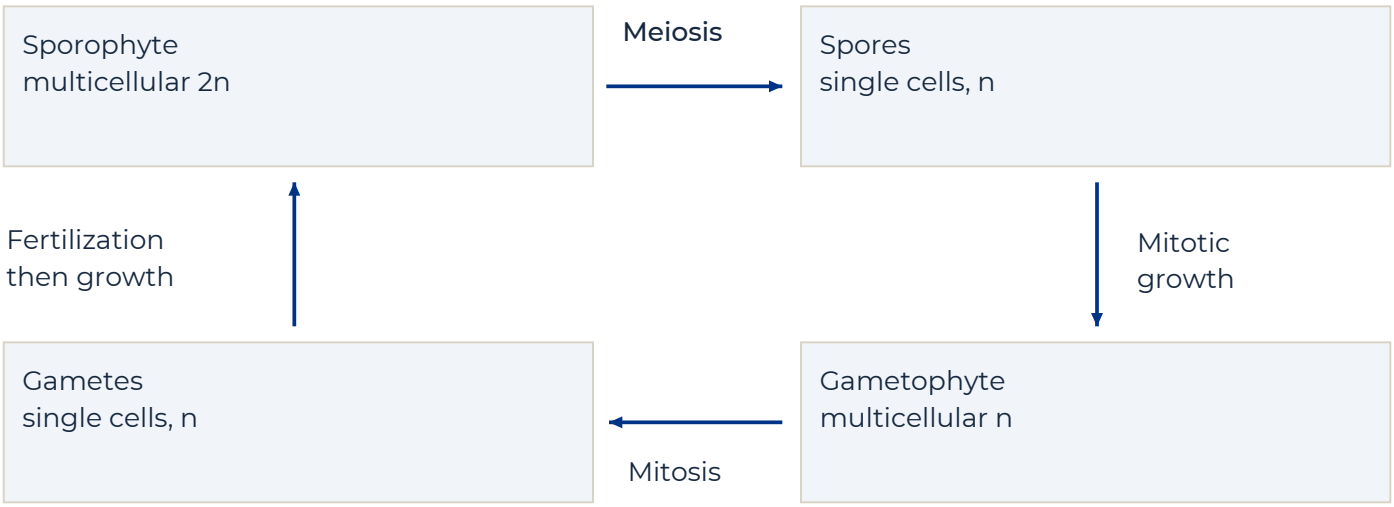
Plant alternation of generations

Track chromosome sets through meiosis, mitosis and fertilization.



If sporophyte 2n = 12	Ploidy	Chromosome count
Spore	n	6
Gametophyte cell	n	6
Gamete	n	6
Zygote / sporophyte cell	2n	12

Spore is not gamete

A spore grows without fusing to another cell. A gamete fuses during fertilization to form the zygote. In the generalized plant cycle, meiosis makes spores; gametophytes make gametes by mitosis.

Dominant generation differs among plant groups. Flowering plants still alternate generations even though their gametophytes are small and dependent. Diagram sizes do not represent duration or physical size.

Plant alternation of generations

Self-check questions and explanations. Refer to the visual guide on the front.

What are the ploidies of the four stages?

Sporophyte 2n; spores n; gametophyte n; gametes n. Sporophyte and gametophyte are multicellular generations; a spore or gamete is a reproductive cell.

Which process connects each stage?

Sporophyte -> spores: meiosis. Spores -> gametophyte: mitotic growth. Gametophyte -> gametes: mitosis. Gametes -> zygote: fertilization; mitotic growth then forms the sporophyte.

How do meiosis, fertilization, spores and gametes differ?

Meiosis halves chromosome-set number; fertilization restores diploidy. A spore grows into a gametophyte without fusion. A gamete fuses with another gamete to form a zygote.

If sporophyte 2n = 12, what counts occur in spore, gametophyte, gamete and zygote?

Spore: n = 6; gametophyte cell: n = 6; gamete: n = 6; zygote: 2n = 12. Count chromosomes by centromeres; DNA replication can change DNA quantity without changing ploidy.

Do plants always make gametes by meiosis?

In the generalized plant cycle meiosis produces haploid spores; haploid gametophytes make gametes by mitosis. In typical animal cycles, meiosis leads to gamete formation without a multicellular haploid generation.

Do tiny gametophytes remove alternation of generations?

No. Both multicellular haploid and diploid generations remain, even if one is microscopic and dependent. In flowering plants, pollen and the embryo sac represent the reduced gametophytes.